r/IGCSE Resources

ATP (Alternative to Practical) Guide to Cambridge IGCSE™

Biology (0610)

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1st edition, for examination until 2025

Version 1

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Drawings

General Rules:

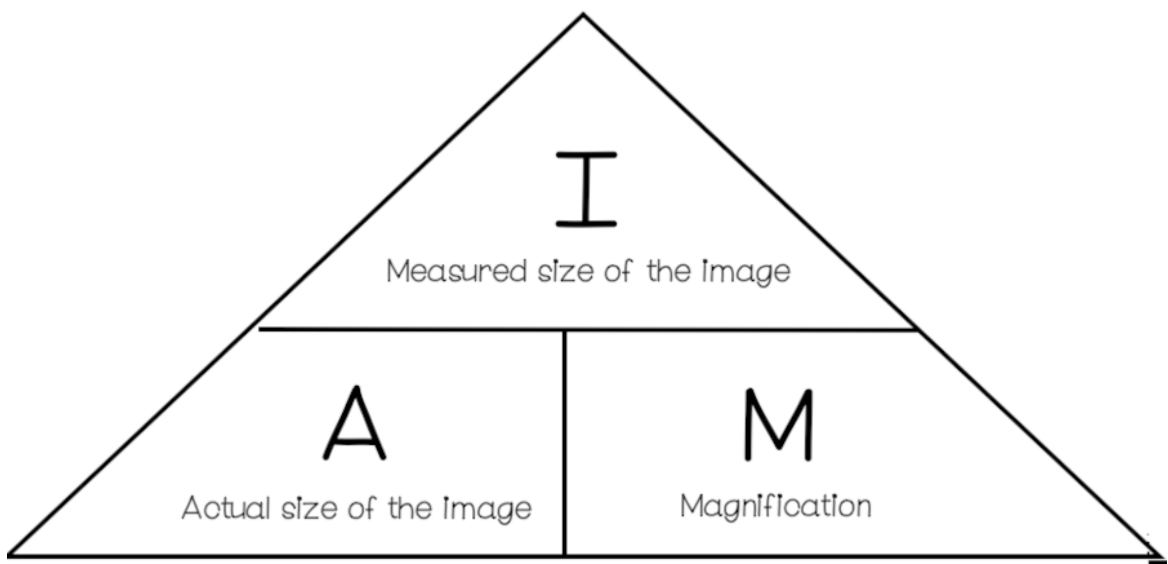
1. Don't draw any details. Try to draw only outlines. (Follow instructions)
2. Take care of any labels that are asked in the question.
3. **NO SHADING !!**
4. Draw the drawing at least 50% larger than the original image provided.
5. Don't use a ruler or a compass to draw, and don't draw broken lines.
6. Read the question carefully to see if any specific segment is required.
7. **Don't draw individual cells !!**
8. Don't draw the surrounding box of the image.

Magnification:

In IGCSE Biology Paper 6, typical drawing and calculation questions always follow. You need to know **the magnification formula**. This Question typically has 2-3 marks, which is pretty simple if you know the formula. (1 mm = 1000 μm)

$$\text{Magnification} = \frac{\text{Image Size}}{\text{Actual Image}}$$

If the image size (μm) and magnification values are given and ask you to find the Actual size (μm), use the Magnification (X) triangle to see the value:



Food Tests

Test for starch:

- Add drops of iodine solution
- Observe colour change
- Positive result: Blue - Black Coloured Solution
- Negative result: Brown Coloured Solution

Test for proteins:

- Add few drops of Biuret reagent
- Observe colour change
- Positive result: Mauve or Purple Coloured Solution
- Negative result: Blue Coloured Solution

Test for Reducing Sugars:

- Add Benedict's solution and heat it
- Observe colour change
- When reducing sugars are detected in a solution, the colour change happens based on the increasing concentration of sugar so blue → green → yellow → orange → red
- Positive result: Brick-Red Coloured Solution
- Negative result: Blue Coloured Solution

Test for fats:

- Known as the Emulsion Test
- Crush the food sample in a mortar with ethanol
- Take a sample in a test tube
- Add drops of distilled water
- Observe a milky-white emulsion showing the presence of fats

Labellings

- Labels should point directly to the image or specific part
- Use a ruler to draw the label line.

Precautions

When doing a experiment planning question in the biology paper 6, you should know all the precautions and how to avoid them:

- Take readings with eye perpendicular to scale (avoid parallax error)
- Take readings from the bottom of the meniscus
- Leave all test tubes in a water bath to equilibrate
- Hot test tube (Use tongs to hold the test tube)
- Stir a solution to distribute all the components evenly
- While taking readings from a measuring cylinder, place it on a flat surface
- Wear a lab coat
- Stand while performing experiment
- Don't expose an alcohol/reagent to a flame directly; heat it using a water bath
- Use burette/pipette instead of measuring cylinder
- In photosynthesis experiment, use a heat shield (beaker of water) placed between the plant and the
- lamp to prevent the temperature from changing due to the changing light intensity
- Heat using water bath (often used in Benedict's test for reducing sugars)
- While using a knife use a solid surface to cut on, cut away from fingers

Sources of errors

- The same type of plant/fruit may not be used; use same species
- Measurement Error (Use a graduated pipette)
- Water loss from tubes was not accounted for; cover the tubes
- Reuse of syringe; (causes contamination) use a new syringe each time
- A subjective/non-quantifiable variable was measured; measure a more quantifiable variable for accurate results
- Subjective Colour Nature (Use a colorimeter)
- No repeats; repeat the experiment 3 times and find the average (to remove anomalies)
- Mass/volume/size/shape of pieces is not the same
- Uneven crushing; use a mortar and a pestle
- Volume of each drop could vary; measure volume used instead of number of drops

- Temperature of water may change; maintain water temperature using a thermostatically controlled water bath
- Heat may be lost through gaps in lid/cork; make sure lid perfectly fits container
- When a variable (eg potato) is cut into smaller pieces, the smaller pieces are not measured into equal sizes
- Number of bubbles counted; measure volume of gas as the volume of each bubble may vary

Hazards

These are the stuff you should always look out for when experiment planning, hazards are not much but they will still get you the point:

- Cutting with a knife may injure hand; cut on a surface and away from hand gloves
- H_20_2 is corrosive; use gloves and goggles while handling it
- While exhaling through a tube into e.g. lime water, there is a risk of lime water being swallowed
- Iodine solution can stain clothes, ensure gloves and safety goggles are worn.

Drawing Graphs

- Use a sharp pencil
- Label both axes with units (IV on the x-axis, DV on the y-axis)
- Labels should be in the form “quantity/units”
- Choose the appropriate scale
- Use crosses “(×)” or “⊙” to mark the data points (for scatter graphs)
- Graph should cover at least half the grid
- Draw a line of best fit that can fit through most of the points as possible (for scatter graphs)
- Join the points (for line graph)

Designing an Experiment

Coming to the final part of this booklet, **how do you design an experiment?** The Paper would consist of an example, and they would ask you to design an experiment based on the example provided. Combining everything you have learned above, I will show you how to do an experiment planning question.

General Information:

Detail	Marks Allotted
Independent Variable	1
Details of Method given/new method	1
Constant Variable	2
Safety Precautions	1
Dependent Variable	1

Method, Controls & their uses:

Method	Use
Gas Syringe or Delivery Tube	Measuring the volume of liquids
Ruler to measure the length of foam produced	Measuring the rate of reaction by measuring the volume of product made.
Cutting plant stem sections every specified amount of time	Measuring the rate of water uptake every set amount of time
Counting the number of bubbles produced per unit time	Measure the rate of reaction by measuring the volume of gas produced
Measure the mass of a potato stick before and after putting in into a solution	Measuring the volume of water that is lost and gained per unit time
Measuring the height a piece of dough has risen	Measuring the rate of reaction in the piece of dough by measuring its change in height

Controls	Method
Temperature	Thermostatically regulated water bath
pH of the Solution	Using a buffer solution

Volume of Solution	Measuring Cylinder or Gas Syringe
Size of Leaves	Square Grid
Time	Timer or Stopwatch
Concentration	Same concentration of (named solution)
Same diameter or size of test tube	-

Additional Tips:

- Stage 3 or more variations for the independent variable (Eg - 5 Concentrations, 5 Species, 5 Acids).
- You can write either in a paragraph or bullet point form.
- Don't spend more than seven minutes on the question.

Sample Question (6 marks)

Tips:

- Use this mnemonic: **I Don't Care So Run Away (IDCSRA)**
- IV – State the independent variable, describe how you are going to change it (include names of apparatus), State the values with units
- DV – State the dependent variable, State the units, State how you are going to measure it
- CV – State three different control variables, describe how you are going to control them
- Safety – State a safety measure and how would you eliminate the risk
- Repeat – State Repeat the experiment three times
- Average – State that you would calculate an average (Numerical Data)

An athlete suggested the hypothesis:

'Drinking a greater volume of beetroot juice would increase the length of time that athletes are able to run.'

Plan an investigation to test this hypothesis.

Exemplar Answer:

- Prepare $20cm^3$, $40cm^3$,$60cm^3$,$80cm^3$ and $100cm^3$ of beetroot juice.

- Consume the beetroot juice at 2 PM everyday
- Measure the length of time the athlete can run at their maximum speed
- Concentration of beetroot juice
- Age, Sex, the diet of athlete
- Same Environment or Conditions of Exercise
- Repeat with the same individuals for many trials
- Medical Staff should be present

Marking Scheme:

1 at least 2 different volumes of beetroot juice ;
2 details of when juice is consumed / ref to fasting / AW ;
3 details of method to measure running time ;
4 rest breaks between repeat measurements ;
5, 6 and 7 named constant variables ;;;
e.g.
- concentration of beetroot juice
- correct detail of consistency in running
- age / stature / sex / fitness, diet, health, of athletes
- same (named) environment conditions of exercise

8 many participants in each group
OR
repeat (whole experiment) with same individual(s) for many trials ;
9 safety precaution ;

Common Independent Variables

Independent Variable	How to change it?
Temperature	Use a thermostatically-controlled water bath and a thermometer to measure the temperature. **Possible Range: 10’C, 20’C, 30’C, 40’C, 50’C, 60’C**
Humidity	With or without plastic bag
Light Intensity	Put a lamp at different distances away, measured using a rule **Possible Range: 10cm, 20cm, 30cm, 40cm, 50cm, 60cm, 70cm**
Air Flow	Use fan or don’t use a fan (with fan = more air flow)
Carbon Dioxide Concentration	Use different concentrations of Sodium Hydrogen carbonate solution that is a source of CO2

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Acknowledgements and Information: